GRIDZILLA

The grid operator's monster. Every house keeps it on a leash.

One simple message from the grid operator to home batteries, cars and heat pumps: how much the house may take or give, and for how long.

An AI runs the grid, a real plant in Blatnica follows it live.

Mathew Stafurik · Peter Mikeška · Matthew · Grid-Serving Storage track · github.com/mathewstars/Gridzilla · gridzilla.victronlab.sk

Europe's largest battery is in people's basements. The grid cannot use any of it

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signals from a grid operator to home batteries in Slovakia today. The one pilot helped the national grid, not the street

4.2 kW

Germany's best case: the operator may dim a new charger down to this. Charging only, used in under 1 % of hours

3 GWh

of home batteries installed in Ukraine in 2025 alone. People charge them by hand before the published outage schedule

What breaks

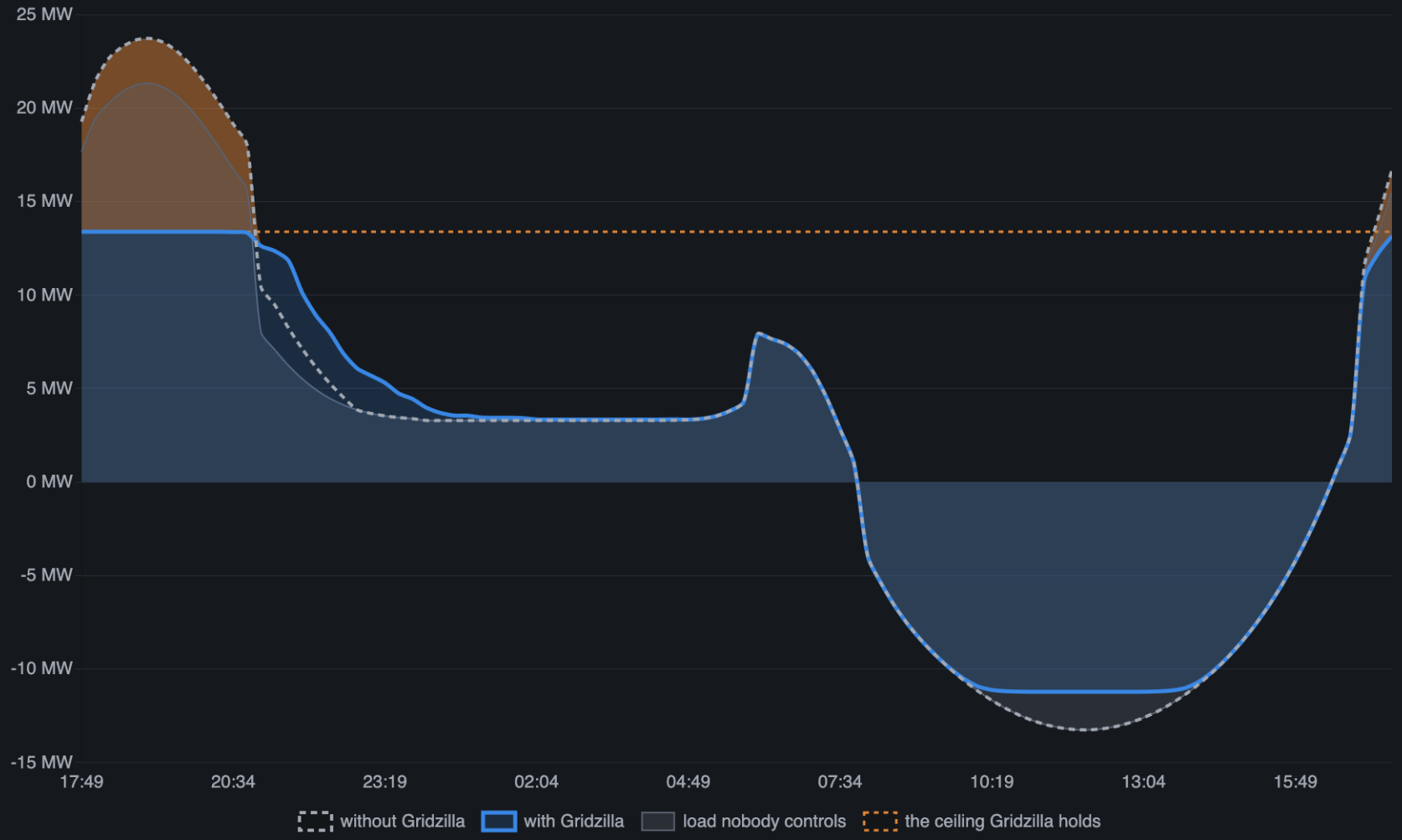
- Dinner time: heat pumps and car chargers push the street transformer past what it can carry; the operator can only switch people off or dig up the street for a bigger one
- Sunny noon: everyone exports at once, the voltage rises, new solar connections are refused or capped for a year
- Crisis: when a transformer is hit or the country is short, everyone is cut equally, including the houses that could have ridden it out

One signed message from the grid operator to every house, on whatever path the maker of the box prefers



- The message names a limit for the whole house: take at most, give at most, give at least, be full by. It never names a device or a setting
- The owner's own limits always win: battery reserve, charger current, opt-out. The house may answer partly, or no
- Every message expires by itself, on the platform and again in the house. If the operator falls silent, the house is back to normal within two minutes
- Paid on the meter: did the house stay under the limit in each 15-minute interval, yes or no, on the operator's own smart meter. Nothing to game

The monster eats peaks. Every street, every evening



79 → 45 %

Vienna's evening peak, share of transformer capacity: 23.7 → 13.4 MW, planned for tonight

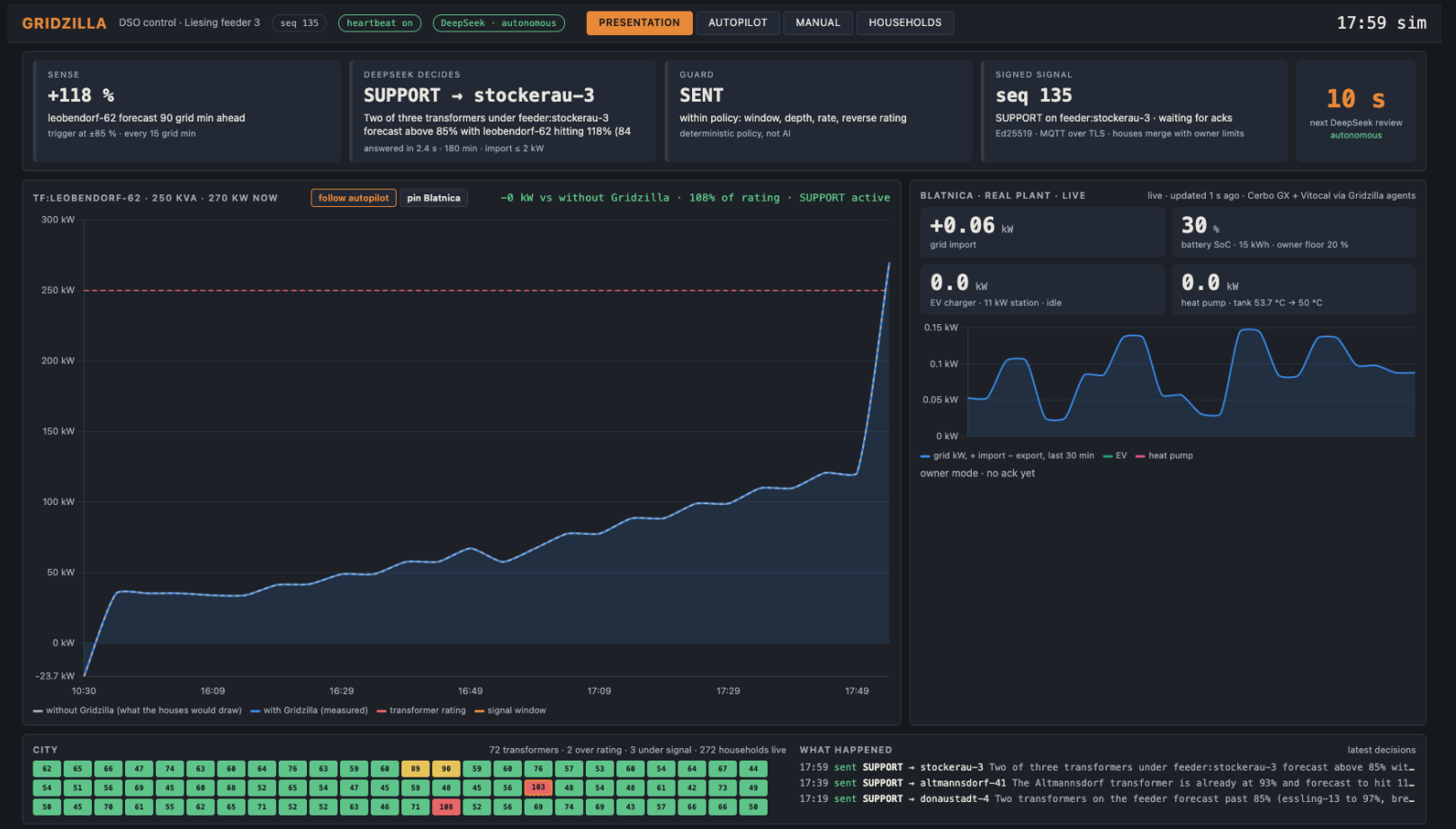
10.5 MWh

of car charging moved from dinner time into the night. Every car full by seven

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transformer upgrades. Batteries eat the dinner spike, the noon sun charges them, nobody at home notices

Live: the Gridzilla board, one real plant inside a simulated Vienna



Ninety seconds, alt-tab

- Board: Vienna, 72 street transformers, 272 households live, one of them real; with and without Gridzilla
- Board: the AI's last decision: what it saw, what it chose, the rule check, the signed message
- Board, Flattening tab: tonight's plan for Vienna, then for one street
- Board, pin Blatnica: the real plant live, last reply verified
- Board: pause the operator's heartbeat, every house goes back to normal

What is real, what is simulated, and what the numbers mean

REAL TODAY

- The whole path: a signed message goes out, a Victron plant and a Viessmann heat pump react, verified replies come back
- The planning and the rule check, running on the fleet every two hours
- Measured on streets with cars: 18 to 31 % off the evening peak, in line with published smart-charging and home-battery trials (3 to 6 kW per car, 1 to 2 kW per battery at the peak)

SIMULATED

- The fleet: 271 simulated households, each standing for 8 real ones: 2,184 houses with a 10 to 25 kWh battery, 1,190 of them with an 11 kW charger needing 6 to 26 kWh a night
- The "without" curve is the same houses left alone: a car charges on 55 % of nights, people come home over three hours, full power until full. Nothing inflated
- These houses carry about 30 % of a street's evening load: a 2030 picture, not a street today

−176 kW

one street, Speising-22: 574 → 398 kW at the evening peak, 31 %, measured tonight

−5.2 MW

Vienna with 2,184 enrolled houses: 26.6 → 21.7 MW at the evening peak, 19 %, measured tonight

0 kW

on a street where no car charged tonight: the gap is the cars and the batteries, nothing else

1

real house in the fleet tonight. The fleet is the part we are asking you to build

NEXT

A pilot street, a second maker, a grid operator that wants the signal

Sign-up tied to the meter and the operator's login: two weeks. A GoodWe version on our own plant: one week.

A ten-house pilot with a Slovak or Austrian grid operator on one real street transformer, this winter.

The vision: streets that keep their lights on from their own batteries when the line is gone, sharing fairly by the same signal.

github.com/mathewstars/Gridzilla · gridzilla.victronlab.sk · protocol, platform, agents and this deck are in the repo

Six signals, one grammar: limits for a window

PREPARE

Fill the battery, the car and the tank to a target by a deadline, in the quiet hours

ABSORB

PV surplus: cap export, charge, soak it into the car and the hot water, curtail last

SUPPORT

Peak or deficit: "take 3 kW only", and if the street needs more, "give at least 2 kW"

STRETCH · vision

Street is dark: grid-forming inverters island the street, batteries are rationed until restoration, export locked to zero. Today's home inverters cannot form a grid; the signal is ready for the ones that will

RECOVER

After restoration: staggered recharge so the street does not refill at once

RELEASE

Event over. Owner mode, immediately

Envelopes: a SUPPORT or ABSORB carries one ceiling per 15-minute slot, the meter's settlement interval. Cohorts let one signal address half a street, so the other half exports while the first charges.

The household is the unit: everything behind one meter, one agent, eight functions

Battery

the lever of last resort: discharge to cover the house or feed the street, within the owner floor

EV charger

the first lever: throttled 16 to 6 A under a cap, soaks surplus, paused when the street is dark

Heat pump

second lever: tank to 55 °C under surplus, floor at 40 °C under a cap, 3 kW shed per house

Boiler

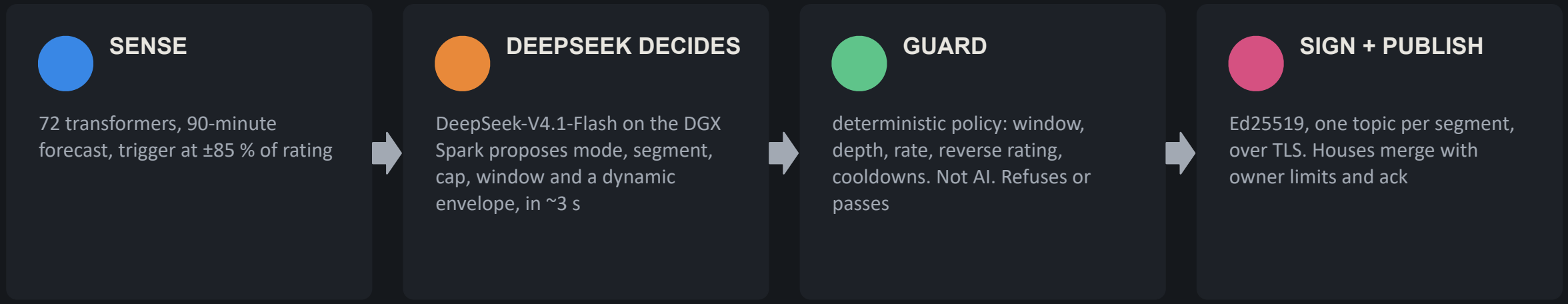
same rules at 65 and 45 °C; an electric boiler is a kilowatt-hour sponge

The agent, any language, forty lines

1. Pin the platform key at registration
2. Subscribe to the segment topics, outbound only
3. Verify signature, sequence, freshness, scope
4. Merge with owner limits: floor, caps, opt-out
5. Apply: the only vendor-specific step
6. Ack, signed with the device key
7. Watch the heartbeat and the window
8. Revert to owner mode

Blatnica, the real one: Victron Cerbo GX with MultiPlus-II and a 15 kWh battery, an 11 kW Victron EV station, and a Viessmann Vitocal heat pump via ViCare. One house on the board.

DeepSeek runs the grid every 15 minutes. The guard is the brake



- Three modes: autonomous, operator approves, off. In autonomous mode nothing waits for a human; the guard and the device-side bounds are the safety
- Scope is physics: a street transformer is relieved only by its own houses; a feeder or substation by every house under it; a village can feed its neighbours through the 22 kV line
- Every decision is logged with its reasoning, latency, verdict and the acks it earned. That log is the audit trail

Everything on that screen runs on real infrastructure today

272

households on the board, 8,070 houses represented

72

street transformers, 24 feeders, 8 substations with Wiener Netze names

2

real devices: a Victron plant and a Viessmann heat pump, 60 km from here

~3 s

DeepSeek decision latency on the DGX Spark

Real plant

Cerbo GX on Venus OS Large, a 53-node Node-RED agent we deploy over the admin API; EV Charging Station throttled first; ESS limits written and restored

Heat pump

Vitocal CU401B through the ViCare REST API with a refresh token; hot-water setpoint and one-time charge, verified on the unit

Transport

Mosquitto on our own server, Let's Encrypt, TLS only, credentials and topic access lists; devices open outbound connections only

Platform

FastAPI on a Ubuntu box behind Apache and Let's Encrypt; registry tree, guard, signing, history, autopilot; public board at gridzilla.victronlab.sk

Simulation

262 households with battery, PV, EV commutes, heat pumps and boilers with real tank physics; each agent speaks the same protocol as the Cerbo

Protocol

Frozen as PROTOCOL.md with JSON Schemas for envelope, signal and ack; any vendor can implement the eight functions from it

Any vendor, four lines each. Any household, paid on the meter

Three paths to the same agent

- Vendor cloud: GoodWe SEMS, Huawei FusionSolar, SolarEdge, Tesla Fleet, SMA, Fronius. Export limit, discharge to cover load, forced discharge schedule, grid charge schedule
- Local controller: Home Assistant, EVCC, OpenEMS, SunSpec Modbus. What the demo runs on Node-RED today
- Standards gateway: EEBus LPC and LPP are literally our import and export caps; OpenADR 3, IEC 61850, OCPP 2.0.1 for chargers
- GoodWe is one apply function away: the registers and the hardware are on the team

Why a DSO pays and a household joins

- Limits are verifiable, reductions are not: the DSO's own smart meter answers yes or no per 15-minute interval. No baseline to inflate
- No ack, no pay; compliance gates the perks; the DSO's existing disconnect is the backstop in a crisis
- The hook: a dynamic export limit beats today's static cap on all but a few afternoons, and gets a refused connection approved
- Plus an availability fee, per-kWh payment for forced feed-in, a full battery before an outage, and a shorter shedding queue
- Noon export is worthless when everyone exports; the same kilowatt-hours leave at six, paid

Production at a DSO: the effort, the cost, the saving

EFFORT

- Protocol and agents are the small part; they exist
- The DSO must add: enrolment bound to the meter point and operator login, a PKI with key rotation, settlement on meter data, the real network model feeding the tree and ratings, 24/7 operations, NIS2 and energy-law compliance
- 10-house pilot on one transformer: 3 months
- A district of a few thousand homes: 12-18 months, 6-8 people, roughly 1.5-3 M€ including the security audit

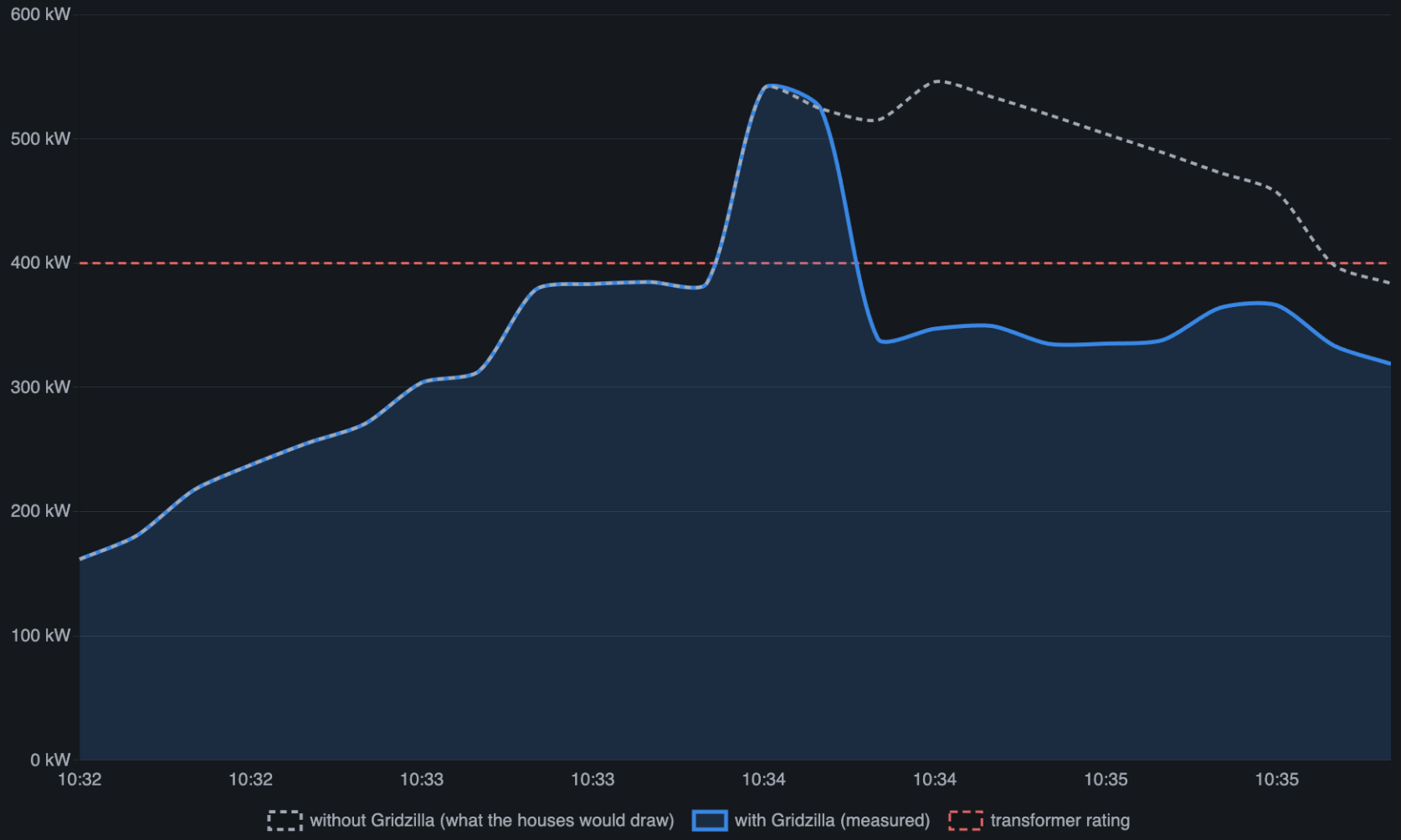
COST TO RUN

- The platform is cheap: two servers and a broker
- The real cost is the incentive: a flat 100-200 € per enrolled house per year, plus energy at crisis prices during STRETCH and SUPPORT
- Paid only to houses whose meter proves they followed the signal. No enforcement, no payment, no saving

SAVING

- Street transformer upgrade 50-150 k€, feeder reinforcement hundreds of k€, a substation tens of M€
- Every street held at ~70 % at the evening peak defers the reinforcement the EV and heat-pump wave otherwise forces this decade
- Vienna scale: hundreds of M€ of capex deferred against a few M€ a year of incentives
- Plus a tool for the outage: PREPARE before it, RECOVER after it, and with grid-forming inverters STRETCH keeps a dark street alive from its own batteries

What the flattening looks like: speising-22, 400 kVA, from an earlier run



−199 kW

largest cut below the counterfactual on this transformer

2 kW

import cap per house DeepSeek chose; EVs throttled first, batteries second

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owner floors crossed; partial acks where a battery hit its reserve

Security of the control plane: the DSO can ask, it can never command

Authenticity

every signal, heartbeat and ack carries an Ed25519 signature; the device pins the platform key and signs with its own

Replay

monotonic sequence numbers persisted on the device, 120 s freshness, heartbeats with sequence and timestamp

Bounded

limits, never setpoints; four-hour window enforced on the device itself; depth and rate capped by the guard

Owner veto

minimum SoC, max current, opt-out merged on the device; a compromised DSO key moves nothing past them

Fail-safe

window end, heartbeat loss or RELEASE restore the owner's settings; a dead platform leaves every house in owner mode

Transport

TLS with a real certificate, per-role credentials, topic ACLs; a device cannot publish a signal even if it tried

Named before you ask

Enrolment is open, any device id can register a key; production binds it to the meter point with a one-time code. The operator board has no login for the demo; production is SSO plus a two-person rule for feeder-level signals. Neither changes the protocol.